# Geological Diversity and Geological History of Canada

The ground on which we stand is underlayed with bedrock that dates back billions of years to the time when the earth's crust first cooled. Some parts of Canada are extremely old and other parts are relatively young when referred to in geological time. The earth's crust has undergone constant folding and faulting where tectonic plates have either collided with each other or pulled apart causing massive change over extremely long periods of time. New mountains and valleys are forming in some regions while at the same time old mountains are being worn down and valleys are being filled in other parts of the country. Most of these changes take millions of years to happen. However, some small changes are immediate as a result of geological events such as earthquakes and volcanoes. Examples of more recent geological events are earthquakes in India in 2001 and 1993, along the San Andreas Fault in San Francisco in 1989 and in Alaska in 1964, and the eruption of Mount St. Helens in 1980. The volcano Kilauea in the Hawaiian chain of islands remains active and constantly growing at a rate of about 400,000 cubic meters a day. In a period of two years it added approximately 200 hectares of new land to the island.

The landmass of Canada, as we know it, has evolved over 4.6 billion years. The geological period from the formation of the earth to 570 million years ago is known as the *Precambrian era*. Rock formed when the earth's crust cooled and broke up into *tectonic plates*. The portion of this original crust that was to become Canada is called the *Canadian Shield* and is the largest landform region in Canada. Over millions of years to follow, the rest of Canada formed around this core.

The *Paleozoic era* followed the Precambrian and lasted about 325 million years. During this time weathering and erosion wore down the Precambrian shields and produced great layers of sediment, which were washed down and deposited into the shallow seas surrounding them. These sediments, over millions of years, were compressed into sedimentary rock that formed a great part of the present day landmass of Canada. Evidence of these geological processes is observed in the occurrence and location of many of our natural resources. The organisms that were trapped in these sedimentary rocks became the oil and gas deposits found in western Canada. Plants that grew along the shores of these ancient seas were trapped in this sediment and formed the coal seams underlying Nova Scotia. As well, these shallow seas were rich in salt minerals and when the seas receded and evaporated, they left behind the salt beds of southwestern Ontario.

Geologists believe that during this era the North American, European and African continental tectonic plates collided to form the *Appalachian Mountains* bordering the shores of Eastern Canada.

The 180 million years following the Paleozoic era is known as the *Mesozoic* or middle era. The *Coast Range Mountains* and the *Innuitian Mountains* were formed during this period when the North American plate split from the European and African and collided on the opposite coast with the Pacific plate. The rift left between these plates filled with massive quantities of magma, which rose from beneath the earth's crust and solidified into igneous rock. As the tectonic plates continued to push against each other, this massive rock bed lifted and folded to form the Coastal Mountains. Toward the end of this era, the continuous forces exerted by the tectonic plates caused the Rocky Mountains to form. In the depression between the mountains ranges of the north, west and east, large swamps existed which were rich in vegetation. These swamps were eventually covered with sand and silt and as the sedimentary layers grew, they were compressed into the sedimentary coal bearing rock of southern British Columbia, Alberta and Saskatchewan.

The Precambrian, Paleozoic and Mesozoic eras bring us to within 66 million years of the present day. This final era of our time is called the *Cenozoic era*. The Rocky Mountains and Coastal Mountains continued to lift and fold during this period. *Volcanoes* erupted and filled many valleys with lava, which formed *plateaus* between the mountain ranges. The seas that covered the central part of our continent continued to recede as the land rose.

It is during the last two million years of the Cenozoic era that the *Ice Age* occurred. During that time, the earth's climate cooled slightly causing large bodies of water to freeze. These massive ice formations grew to hundreds of meters with precipitation that came in the form of snow and sleet and formed *glaciers*, which appeared to advance as they grew and spread. With slight increases in climatic temperature the glaciers would melt and recede or shrink. This continuous back and forth motion of the thick ice sheets that were embedded with rocks and sediment scoured the rock surfaces and wore away mountaintops. The melting ice caused erosion and deposition of sand, silt and rock debris at the advancing edges of the glaciers and into the lowlands. Evidence of these actions can be seen in the form of drumlins, moraines, eskers and raised beaches. It is thought that there were several periods of glacial activity with the last being about 6000 years ago. Remnants of the Ice Age can still be found in the glaciers of the Arctic and in the higher elevations of the Rocky Mountains.

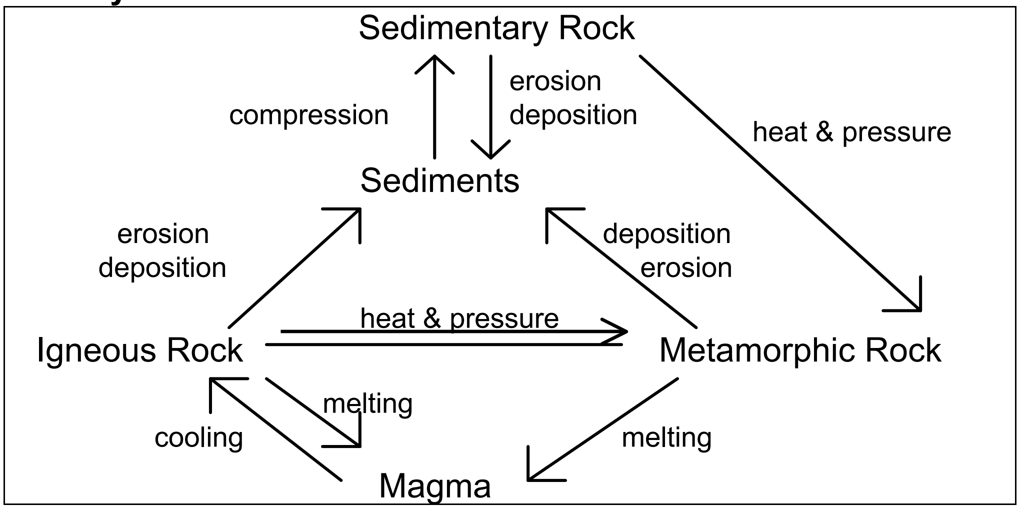
**Table 1: Ecozone percentages for each Rock Age and combined Rock Ages**

| **Rock Age** | **Cenozoic** | **Mesozoic** | Mes-Cen | **Paleozoic** | Pal-Mes | **Precambrian** | Prec-Mes | Prec-Pal | Unknown |
|---|---|---|---|---|---|---|---|---|---|
| Arctic Cordillera | 0.30 | 7.96 | 0.29 | 22.25 | 0.00 | 54.80 | 0.00 | 1.49 | 12.86 |
| Northern Arctic | 2.11 | 6.41 | 2.77 | 40.26 | 0.00 | 47.42 | 0.00 | 0.28 | 0.59 |
| Southern Arctic | 3.38 | 6.36 | 0.00 | 12.56 | 0.00 | 77.68 | 0.00 | 0.01 | 0.00 |
| Taiga Plains | 1.39 | 51.22 | 0.49 | 43.78 | 0.00 | 3.11 | 0.00 | 0.02 | 0.00 |
| Taiga Shield | 0.05 | 0.00 | 0.00 | 0.64 | 0.00 | 99.31 | 0.00 | 0.00 | 0.00 |
| Boreal Shield | 0.00 | 0.21 | 0.00 | 5.05 | 0.00 | 94.22 | 0.00 | 0.30 | 0.08 |
| Atlantic Maritime | 0.00 | 1.20 | 0.00 | 93.29 | 0.02 | 2.51 | 0.00 | 2.46 | 0.51 |
| Mixedwood Plains | 0.00 | 0.12 | 0.00 | 94.52 | 0.00 | 4.94 | 0.00 | 0.27 | 0.14 |
| Boreal Plains | 6.74 | 70.25 | 0.00 | 20.93 | 0.00 | 2.07 | 0.00 | 0.00 | 0.00 |
| Prairies | 10.81 | 85.01 | 0.81 | 3.37 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| Taiga Cordillera | 2.90 | 20.22 | 0.06 | 47.04 | 0.08 | 14.56 | 0.00 | 15.13 | 0.00 |
| Boreal Cordillera | 6.65 | 31.79 | 0.05 | 33.81 | 4.85 | 4.80 | 5.01 | 11.97 | 1.07 |
| Pacific Maritime | 14.10 | 67.47 | 1.70 | 5.30 | 0.59 | 3.81 | 0.00 | 0.24 | 6.79 |
| Montane Cordillera | 17.91 | 40.55 | 0.40 | 18.98 | 3.34 | 12.80 | 0.00 | 4.68 | 1.32 |
| Hudson Plains | 0.00 | 2.05 | 0.00 | 80.63 | 0.00 | 17.32 | 0.00 | 0.00 | 0.00 |

Note that this table has the four major rock age classifications. But also note that there are columns with a combination of rock age classes, e.g., Mes-Cen, or Mesozoic and Cenozoic. This means that although the majority of the rock surface that is exposed is of one of the four classes, there are some areas that show a combination of two types.

There are three main classes of rock: *igneous*, *metamorphic* and *sedimentary* rock. Hot, melted *magma* from beneath the earth's crust that is forced to the surface, cools and hardens and forms igneous rock. When extreme heat and pressure are applied to this rock, for example as friction between tectonic plates, the rock is transformed into metamorphic rock. As natural forces wear down the igneous rock, it becomes sediments. If these sediments undergo extreme compression, they are transformed into sedimentary rock. It is a constant loop cycle between these three forms of rock.

## The Rock Cycle




The next table deals with the types of rock found in Canada divided by Ecozone.

**Table 2: Ecozone percentages for each Rock Type and combined Rock Types**

| **Rock type** | **Igneous** | **Metamorphic** | **Sedimentary** | Sed-Ign | Unknown |
|---|---|---|---|---|---|
| Arctic Cordillera | 11.49 | 23.72 | 47.99 | 3.90 | 12.86 |
| Northern Arctic | 11.64 | 24.59 | 60.65 | 2.36 | 0.59 |
| Southern Arctic | 40.17 | 16.71 | 40.93 | 2.17 | 0.00 |
| Taiga Plains | 0.69 | 0.00 | 99.32 | 0.00 | 0.00 |
| Taiga Shield | 55.95 | 28.40 | 15.47 | 0.18 | 0.00 |
| Boreal Shield | 54.50 | 28.63 | 15.88 | 0.77 | 0.08 |
| Atlantic Maritime | 17.26 | 0.74 | 79.87 | 1.62 | 0.51 |
| Mixedwood Plains | 1.16 | 3.23 | 95.46 | 0.00 | 0.14 |
| Boreal Plains | 1.20 | 0.65 | 98.09 | 0.06 | 0.00 |
| Prairies | 0.00 | 0.00 | 100.00 | 0.00 | 0.00 |
| Taiga Cordillera | 3.87 | 0.00 | 96.12 | 0.00 | 0.00 |
| Boreal Cordillera | 32.55 | 1.07 | 65.35 | 1.03 | 0.00 |
| Pacific Maritime | 70.57 | 6.28 | 18.68 | 0.03 | 4.44 |
| Montane Cordillera | 38.10 | 2.81 | 57.12 | 1.96 | 0.00 |
| Hudson Plains | 11.40 | 4.51 | 84.03 | 0.06 | 0.00 |

Canada has abundant mineral resources within these rocks. We are familiar with the metals: iron ore, aluminum, copper, nickel, lead, uranium, zinc, gold and silver. But Canada is also a producer of lesser-known metals such as antimony, bismuth, cadmium, cesium, cobalt, lithium, magnesium, molybdenum, selenium and tellurium. Non-metals are also important to our economy. These include gems (such as amethyst, diamonds, garnet, jade and opal), manufacturing raw materials (such as asbestos, graphite, gypsum, mica and pumice), and agricultural additives (such as peat, potash and potassium sulfate).

The bedrock also provides us with building materials such as granite, limestone, marble, sandstone, shale and slate.